

AQUATIC ENVIRONMENTS

Planet Earth is really planet Ocean

More than 70 percent of the Earth's surface is covered by water, which in its liquid form is essential for life. Water is continually circulating around the planet, evaporating from its surface and being carried as water vapor in the atmosphere until it falls again as rain. Around 95 percent of the Earth's water is salt water, which is found in seas, oceans, and coastal lagoons as well as a few isolated soda and salt lakes. The other 5 percent is fresh water, which is seen in rivers and lakes, but also includes the ice held in polar regions and glaciers, and groundwater that is hidden from view. The challenges of life in fresh and salt water are very different and relatively few animal species are able to move from one to the other.

Rivers and lakes

Fresh water is vital for life on land—without water, plants cannot grow and animals would have nothing to drink. Rivers and lakes create diverse habitats

ranging from fast-flowing rivers and swampy wetlands to the relatively calmer but deeper water of many lakes. Organisms living in lakes and rivers must cope with strong currents, survive freezing conditions in winter, and endure summer droughts when some rivers and



LAKES

Lakes are often isolated, with little opportunity for new aquatic species to colonize them (unless introduced by humans). As a result they may have large numbers of endemic species or subspecies that have evolved to exploit the available habitats.



MANGROVES

These wetlands provide safe inshore nurseries for various marine animals as well as breeding sites and roosts for many bird species, including the scarlet ibis. These wading birds use their long, curved, sensitive bills to locate food in the soft mud.



RIVERS

The steeper the gradient of a river, the faster the water flows and the stronger the current. More animal species tend to be found downstream, where slower-flowing water allows aquatic plants to grow. This increases the number of habitats.

CALIFORNIA COAST ▸

The sheer power of a breaking wave in Monterey Bay, CA, is revealed.

